Sec 5.1 #8

Thm. Let $V$ be a finite vector space, $T: V \to V$ be a linear operator.

If $\beta, \gamma$ are distinct ordered basis of $V$,

then $\det([T]_\beta) = \det([T]_\gamma)$.

Proof. Since

$$\det([T]_\beta) = \underbrace{\det([T]_\beta^\gamma)} \cdot \det([T]_\beta^\beta) \cdot \underbrace{\det([T]_\gamma^\beta)}_{\text{inverse}}$$

$$= \det([T]_\beta^\gamma [T]_\beta^\beta [T]_\gamma^\beta) = \det([T]_\gamma^\gamma)$$

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "\phi_\alpha"'] \arrow[dr, phantom, "G"] & V \arrow[d, "\phi_\beta"] \\ F^n \arrow[r, "{[T]_\alpha^\beta}"'] & F^n \end{tikzcd}$$

Sec 5.1 #9

Thm. Let $T: V \to V$ be a linear operator on finite, v.s $V$.

$T$ is invertible iff every eigenvalue of $T$ is nonzero.

Proof. There exists a zero eigenvalue

$\Leftrightarrow \exists v \neq 0$ s.t $T(v) = 0 \cdot v = 0$

$\Leftrightarrow \ker T \neq \{0\}$

$\Leftrightarrow T$ is not injective.

Thm. Let $T: V \to V$ be an invertible linear operator on finite v.s $V$.

$\lambda$ is eigenvalue of $T$ iff $\lambda^{-1}$ is eigenvalue of $T^{-1}$.

Proof. $\lambda$ is eige. of $T$

$\Leftrightarrow \exists v \neq 0$ s.t $T(v) = \lambda v$

And, $T(v) = \lambda v$

$\Leftrightarrow (T^{-1} \circ T)(v) = T^{-1}(\lambda v) = \lambda \cdot T^{-1}(v)$

$\Leftrightarrow \lambda^{-1} v = T^{-1}(v)$.

Sec 5.1 11212

observation: Let $V$ be fin. v.s. $I_V: V \to V$ be a identity map.

$$\det(\lambda I_V - tI_V) = \det((\lambda - t)I) = (\lambda - t)^n$$

Thm. If $A$ is diagonalizable Matrix with a unique eigenvalue, then $A = \lambda I$ where $\lambda$ is eigenvalue of $A$

proof. since $A$ is diagonalizable, and has only one eigenvalue,

$$A = Q^{-1} D Q \quad \text{where} \quad D = \lambda I$$
$$= Q^{-1} Q \lambda I$$
$$= \lambda I$$

example

$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is <u>not</u> diagonalizable;

$\det \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = (1-t)^2$, thus has eigenvalue as only 1.

But, if $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is diagonalizable, it is scalar Matrix.

This is Contradiction.

Thm. If a matrix $A$ is similar to $B$, then $\det A = \det B$.

Proof. $\det A = \det Q^{-1} B Q = \det Q^{-1} \det Q \det B = \det B$.

Thm. charicteristic of linear operator does not depend on a basis.

Proof. Since $A = Q^{-1} B Q \Rightarrow A - I = Q^{-1} B Q - I$
$$= Q^{-1}(B - I) Q$$

Sec 5.1. #14

Thm. Let $T: V \to V$ be a linear operator on finite vec. sp. over $F$.

Put $A = [T]_\beta$.

1. For $v \in V$, if $\phi_\beta(v)$ is eigenvector of $A$ corresponding to $\lambda$, then $v$ is eigenvector of $T$ corresponding to $\lambda$

2. If $\lambda \in F$ is eigenvalue of $A$ and $T$, then

$y \in F^n$ is eigenvector of $A$ corresponding to $\lambda$ iff $\phi_\beta^{-1}(y)$ is eigenvector of $T$ corre- to $\lambda$.

Proof. 1.

$$\phi_\beta(T(v)) = [T(v)]_\beta = [T]_\beta [v]_\beta = A \cdot \phi_\beta(v) = \lambda \phi_\beta(v) = \phi_\beta(\lambda v)$$

Since $\phi_\beta$ is isomorphism, $T(v) = \lambda v$.

2.

$$\lambda \phi_\beta^{-1}(y) = \phi_\beta^{-1}(\lambda y) = \phi_\beta^{-1}(Ay) = \phi_\beta^{-1}([T]_\beta y) = \phi_\beta^{-1}([T]_\beta [\phi_\beta^{-1}(y)]_\beta)$$

$$= \phi_\beta^{-1}([T(\phi_\beta^{-1}(y))]_\beta)$$

$$= T(\phi_\beta^{-1}(y))$$

Thm. $(\lambda, x)$ is eigen- of $T: V \to V$. Then for all $n \in \mathbb{N}$, $(\lambda^m, x)$ is eigen- of $T^m$.

Proof. $T^m(x) = T^{m-1}(\lambda x) = \lambda \cdot T^{m-1}(x) = \cdots = \lambda^m x$.

sec 5.1 #19

Thm. Let $T: V \to V$ linear, $\dim V < \infty$. $c \in F$ be a scalar.

If $\lambda \in F$ is eigenvalue of $T$ corresponding to $v \in V$

then $\lambda - c$ is eigenvalue of $T - cI$ corresponding to $v \in V$. Moreover,

$T$ is diagonalizable iff $T - cI$ diagonalizable.

Proof. $T(v) = \lambda v$ implies

$$(T - cI)(v) = T(v) - cv = (\lambda - c)v.$$

Sec 5.1 #18

Thm. Let $T: M_{n\times n}(\mathbb{R}) \to M_{n\times n}(\mathbb{R}) : A \mapsto A^t$.

1. Eigenvalue of $T$ is only $\pm 1$.
2.

Proof. Let $\lambda \in \mathbb{R}$ be an eigenvalue of $T$ corresponding to $A$. Then,

$$T(A) = \lambda A = A^t$$

$$\lambda A = A^t \iff \lambda A^t = A$$

Thus, $\lambda A = \lambda^{-1} A \Rightarrow (\lambda^2 - 1)A = 0$. Since $A \neq 0$, $\lambda^2 = 1$.

Let $\beta$ be a basis of $M_{n\times n}(\mathbb{R})$.

$[T]_\beta$ is diagonal iff $\forall A \in \beta$, $T(A) = \pm A^t$.

$$\beta = \left\{ \begin{pmatrix} 1 & 0 & \\ 0 & 0 & \\ & & \ddots \; 0 \end{pmatrix}, \begin{pmatrix} 0 & & \\ & 1 & \\ & 0 & \ddots \end{pmatrix}, \cdots, \begin{pmatrix} 0 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix} \right\} \leftarrow n \text{ elements}$$

$$\cup \left\{ \begin{pmatrix} 0 & & 1 \\ 1 & \ddots & \\ & & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & \ddots & \\ 0 & 0 & \ddots & \\ 1 & 0 & & \end{pmatrix}, \cdots \right\} \leftarrow \frac{n(n-1)}{2}$$

$$\cup \left\{ \begin{pmatrix} 0 & -1 & & \\ 1 & 0 & \ddots & 0 \\ & & \ddots & \\ & 0 & & \end{pmatrix}, \cdots \right\}$$

$n^2$ elements

sec 5.1 #19

Thm. $A, B \in M_{n\times n}(\mathbb{C})$.

For an invertible $B$, there exists $c \in \mathbb{C}$ s.t $A+cB$ is not invertible.

Proof. $A+cB$ is not invertible

$\Leftrightarrow \det(A+cB)=0 \Leftrightarrow \det(B^{-1}A+cI)=0$

$\Leftrightarrow -c$ is eigenvalue of $B^{-1}A$.

Since field is set by $\mathbb{C}$, $B^{-1}A$ always has a eigenvalue $\lambda$.

Now, taking $c=-\lambda$.

Put $A=\begin{pmatrix}1&1\\0&1\end{pmatrix}$, $B=\begin{pmatrix}0&0\\1&1\end{pmatrix}$

Thm. Let $A \in M_{n\times n}(F)$, $f(t)=\det(A-tI)$.

Then, $f(0)=\det A$. More precisely,

$$f(t)=(-1)^n t^n + a_{n-1}t^{n-1}+\cdots+a_1 t+\det A.$$

And, for some polynomial $\deg g(t) \le n-2$

$$f(t)=(A_{11}-t)(A_{22}-t)\cdots(A_{nn}-t)+g(t).$$

And, $tr(A)=(-1)^{n-1}a_{n-1}$.

Proof. Using induction.

For $n=2$, $f(t)=(a-t)(d-t)-bc=(A_{11}-t)(A_{22}-t)-bc$

Suppose for $n \in \mathbb{N}$, the statement is true.

Then, for $A=\left(\begin{array}{c|c} B & \\ \hline & A_{n+1,n+1}\end{array}\right)$, $\det(A-tI)=\det(B-tI)\cdot(A_{n+1,n+1}-tI)$

$=$ ~~~~~~~~~~

Now, $f(t)=(A_{11}-t)(A_{22}-t)\cdots(A_{nn}-t)+g(t)$

$=(-1)^n t^n+(-1)^{n-1}(A_{11}+A_{22}+\cdots+A_{nn})t^{n-1}+g(t)+\cdots$

Thm. Let $T: V \to V$ be a linear operator, $g(t) \in F[t]$.

If $x$ is eigenvector of $T$ corresponding to $\lambda$, then $g(T)(x) = g(\lambda)x$.

Proof. $\left(\sum_{i=0}^{n} a_i T^i\right)(x) = \sum_{i=0}^{n} a_i T^i(x) = \sum_{i=0}^{n} a_i \lambda^i$

Thm. If a linear operator $T: V \to V$ is diagonalizable and $f(t)$ is characteristic poly. of $T$, then $f(T) = 0$.

Proof. Let $x \in V$ be given. Then, $x = \sum_{i=1}^{n} a_i x_i$ where $x_i$ is eigenvector corresponding to $\lambda_i$

$$f(T)\left(\sum_{i=1}^{n} a_i x_i\right) \overset{\text{def of sum of functions}}{=} \sum_{i=1}^{n} f(T)(a_i x_i) \overset{T(x)=\lambda x \Rightarrow T(ax)=a\lambda x}{=} \sum_{i=1}^{n} a_i \underbrace{f(\lambda_i)}_{=0}(x_i) = \sum^{n} 0 = 0.$$

Thm. Let $\lambda_1, \lambda_2, \cdots, \lambda_k$ be distinct eigenvalues of $T: V \to V$.
And, for each $1 \le i \le k$, define $S_i$ is a basis of $E_{\lambda_i}$.
Then, $\bigcup_{i=1}^{k} S_i$ is linearly independent.

Proof. Using induction. For $k=1$, trivial.
Suppose that for $n \in \mathbb{N}$, the statement is true.
Denote $S_i = \{v_{i,1}, v_{i,2}, \cdots, v_{i,n_i}\}$. Let

$$\sum_{i=1}^{n+1} \sum_{j=1}^{n_i} a_{ij} \cdot v_{ij} = 0.$$

Taking $T - \lambda_{n+1} I$ for both side, then

$$\left(T - \lambda_{n+1} I\right)\left(\sum_{i=1}^{n+1} \sum_{j=1}^{n_i} a_{ij} \cdot v_{ij}\right)$$

$$= \sum_{i=1}^{n+1} \sum_{j=1}^{n_i} a_{ij} (\lambda_i - \lambda_{n+1}) \cdot v_{ij}$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{n_i} a_{ij} (\lambda_i - \lambda_{n+1}) v_{ij}$$

By assumption, this implies $\forall 1 \le i \le n, \forall 1 \le j \le n_i$, $a_{ij}(\lambda_i - \lambda_{n+1}) = 0$.
Since $\lambda_i \ne \lambda_{n+1}$ $\forall 1 \le i \le n$, $a_{ij} = 0$.
Now, $\sum_{i=1}^{n+1} \sum_{j=1}^{n_i} a_{ij} \cdot v_{ij} = \sum_{j=1}^{n_{n+1}} a_{n+1,j} \cdot v_{n+1,j} =$

Meanwhile, $S_{n+1}$ is linearly independent, $\forall 1 \le j \le n_{n+1}$, $a_{n+1,j} = 0$.

「Thm. Let $T: V \to V$ be a linear map on finite vector space $V$, and $\lambda$ be an eigenvalue with multiplicity $m$. Then,

$$1 \le \dim E_\lambda \le m$$

Proof. Using replacement Theorem, Let basis of $E_\lambda$ as $\{v_1, \cdots, v_p\}$ and basis of $V$ as $\beta = \{v_1, \cdots, v_p, \cdots, v_n\}$

Then,

$$[T]_\beta = \begin{pmatrix} \lambda I_p & B \\ O & C \end{pmatrix}.$$

Now, $f(t) = \det([T]_\beta - tI) = \det \begin{pmatrix} (\lambda - t) I_p & B \\ O & C - tI \end{pmatrix} = \det((\lambda - t) I_p) \cdot \det(C - tI) = (\lambda - t)^p g(t)$

Therefore, $\dim E_\lambda = p \le \deg f(t) = m$

Thm. Let $T: V \to V$ be a linear operator on finite vector space $V$.

$T$ is diagonalizable iff:

1). Characteristic polynomial of $T$ splits over $F$.

2). For all eigenvalue $\lambda$, multiplicity of $\lambda = \dim E_\lambda$.

Proof. Suppose that $T$ is diagonalizable. Put $\beta = \{v_1, \cdots, v_n\}$ be a basis s.t. $[T]_\beta$ is diag.

$T(v_i) = \lambda_i v_i$ for some $\lambda_i \in F$, since $[T]_\beta$ is diagonal.

And, the characteristic polynomial of $T$:

$$f(t) = \prod_{i=1}^{n} (\lambda_i - t).$$

Thus, $\dim E_{\lambda_i} =$ multi. of $\lambda_i$.

Conversely, suppose that

Thm. Let $V$ be a finite vector space, and $W_1, W_2, \cdots, W_k \leq V$ be subspaces.

TFAE:

1) $V = W_1 \oplus W_2 \oplus \cdots \oplus W_k$

2) $V = \sum_{i=1}^{k} W_k$ and if $\sum_{i=1}^{k} V_k = 0$ where $V_i \in W_i$, then $V_1 = V_2 = \cdots = V_k = 0$.

3). For any $v \in V$, there exists a unique representation s.t

$$v = v_1 + \cdots + v_k \quad \text{where } v_i \in W_i.$$

4). For basis $\gamma_i$ of $W_i$, $\bigcup_{i=1}^{k} \gamma_i$ is basis of $V$.

5.) For each $1 \leq i \leq k$, there exist a basis $\gamma_i$ of $W_i$ s.t

$$\bigcup_{i=1}^{k} \gamma_i \text{ is basis of } V.$$

Thm.

A linear operator $T: V \to V$ is diagonalizable

iff

$V$ is direct sum of eigenspaces of $T$.

Thm. Let $T: V \to V$ be a linear operator on finite vector space $V$.
If there exists a basis $\beta$ of $V$ s.t $[T]_\beta$ is upper-triangle Matrix,
then characteristic polynomial of $T$ splits over $F$.

Proof. Using induction. For $\dim V = 2$, trivial.
Suppose that for $\dim V = n \in \mathbb{N}$, the statement is true.
Let $T: F^{n+1} \to F^{n+1}$ be a linear operator, and $\beta$ be a basis of $F^{n+1}$ s.t $[T]_\beta$ is up-tri.
Denote $\beta = \{v_1, \cdots, v_n, v_{n+1}\}$. Then, for a restriction

$$T|_{F^n} : F^n \to F^n ; x \mapsto (\pi_1 \circ T(x), \pi_2 \circ T(x), \cdots, \pi_n \circ T(x))$$

Satisfies: $\left[T|_{F^n}\right]_{\beta \setminus \{v_{n+1}\}}$ is upper-triangular,

since for all $1 \le i \le n$, $(\pi_{n+1} \circ T)(v_i) = 0$, $(T|_{F^n}(v_i), 0) = T(v_i)$.

Now,

$$\det(T - tI) = \det \begin{pmatrix} T|_{F^n} - tI & \vdots \\ 0 \cdots 0 & \lambda - t \end{pmatrix} = (\lambda - t) \cdot \det(T|_{F^n} - tI) = f(t)$$

And, by assumption, $\det(T|_{F^n} - tI)$ splits over $F$, so is $f(t)$.

Thm. Let $T: V \to V$ be a linear operator on $V \cong F^n$,
$\lambda_1, \cdots, \lambda_k$ be distinct eigenvalues with multiplicity $m_i$.
If there exists a basis of $V$ s.t $[T]_\beta$ is upper-triangle,
then

$$[T]_\beta = \begin{pmatrix} \lambda_1 I_{m_1} & & & 0 \\ & \ddots & & \\ & & \lambda_i I_{m_i} & \\ 0 & & & \ddots \; \lambda_k I_{m_k} \end{pmatrix} + \begin{pmatrix} 0 & \ddots & \ddots & \\ & \ddots & 0 & \\ 0 & & \ddots & \\ & & & 0 \end{pmatrix}.$$

proof. Since $[T]_\beta$ is upper-triangle, $\det([T]_\beta - tI) = \prod_{i=1}^{n} (A_{ii} - t)$. where $A = [T]_\beta$.
And, by assumption, $f(t) = (\lambda_1 - t)^{m_1} \cdots (\lambda_k - t)^{m_k} = \prod_{i=1}^{n} (A_{ii} - t)$.
Thus proved.

Thm. Let $A \in M_{n\times n}(F)$ be a matrix which is similar to an upper triangle, and $\lambda_1, \cdots, \lambda_k$ be distinct eigenvalue of $A$ with multiplicity $m_i$.

1) $\operatorname{tr} A = \sum_{i=1}^{k} m_i \lambda_i$

2) $\det A = \prod_{i=1}^{k} \lambda_i^{m_i}$.

Proof. Trivial.

Thm. If a characteristic polynomial of $A \in M_{n\times n}(F)$ splits over $F$, then $A$ is similar to an upper triangle matrix.

Proof. For $n=1$, trivial.

Suppose that for $n-1 \in \mathbb{N}$, the statement is true.

Let $A \in M_{n\times n}(F)$, which is characteristic polynomial splits.

Denote

$$A = \begin{pmatrix} A' & \vdots \\ \cdots & a_n \end{pmatrix}.$$ where $A' \in M_{n-1\times n-1}(F)$.

Since $f_A(t)$ splits over $F$, there is an eigenvector $v_n \in V$.

Let $\beta = \{v_1, \cdots, v_n\}$ be a basis of $F^n$.

Put $P = \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}$. Then,

$$P^{-1}AP = \begin{pmatrix} P^{-1}Av_1 & \cdots & P^{-1}Av_n \end{pmatrix} = \begin{pmatrix} P^{-1}Av_1 & \cdots & P^{-1}\lambda_n v_n \end{pmatrix}$$

$\lfloor$

[Thm. Let $T: V \to V$ be an invertible linear operator. with $\dim V = n$.
If $\lambda \in F$ is eigenvalue of $T$, then $\lambda^{-1}$ is eigenvalue of $T^{-1}$.
Precisely, $\{v \in V \mid T(v) = \lambda v\} = \{v \in V \mid T^{-1}(v) = \lambda^{-1} v\}$.

Moreover, $T$ is diagonalizable iff $T^{-1}$ is diagonalizable.

Proof. Since $T(v) = \lambda v$ for some non-zero vector $v$,

$$T(v) = \lambda v \iff v = (T^{-1} \circ T)(v) = T^{-1}(\lambda v) = \lambda T^{-1}(v)$$
$$\iff T^{-1}(v) = \lambda^{-1} v.$$

Suppose that $T$ is diagonalizable. Then, there exists a basis $\beta$ of $V$ s.t

$$[T]_\beta = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix},$$ which is diagonal.

Now, $$[T^{-1}]_\beta = [T]_\beta^{-1} = \begin{pmatrix} \lambda_1^{-1} & & & 0 \\ & \lambda_2^{-1} & & \\ & & \ddots & \\ 0 & & & \lambda_n^{-1} \end{pmatrix}$$

[Thm. Let $A \in M_{n \times n}(F)$ be given. Then, $A$ and $A^t$ has same characteristic polynomial.
That is, $A$ and $A^t$ has same eigenvalue with same multiplicity.
Denote $E_\lambda$ is eigenspace of $A$ corresponding to $\lambda$,
$E'_\lambda$ is " $A^t$ " $\lambda$.
For any eigenvalue $\lambda$, $\dim E_\lambda = \dim E'_\lambda$.
Moreover, $A$ is diagonalizable iff $A^t$ is diagonalizable.
Proof.

[Def. Let $V$ be a finite vector space, $T, U: V \to V$ be linear operator.

If there exists a basis $\beta$ s.t

$$[T]_\beta, [U]_\beta \text{ are diagonal Matrix.}$$

Then, it is said to be $T$ and $U$ is simultaneously diagonalizable.

In Matrix,

$$\exists Q \in M_{n\times n} \text{ s.t } \begin{cases} Q^{-1}AQ \\ Q^{-1}BQ \end{cases} \text{ are diagonal.}$$

[Thm. If $T, U: V \to V$ are simultaneously diagonalizable,

then $T, U$ is commute. $(TU = UT)$.

proof. Since diagonal Matrices are commute,

$$[T \circ U]_\beta = [T]_\beta [U]_\beta = [U]_\beta [T]_\beta = [U \circ T]_\beta.$$

[Thm. If $T$ is diagonalizable,

then for all $n \in \mathbb{N}$, $T, T^n$ are simul-diagonal-.

proof. $[T^n]_\beta = [T]_\beta^n$ is diagonal, especially

$$= \begin{pmatrix} \lambda_1^n & & 0 \\ & \lambda_2^n & \\ 0 & & \ddots \ \lambda_k^n \end{pmatrix}.$$

Thm. Let $T: V \to V$ be a linear operator, and $W \leq V$.

$W$ is called $T$-invariant subspace of $V$ if:

$$T[W] \subseteq W.$$

And, for $x \in V$,

$$W = \langle \{ T^n(x) \mid n = 0, 1, \cdots \} \rangle$$

is called $T$-cyclic subspace generated by $x$.

Thm. Let $T: V \to V$ be a linear operator on fin.v.s $V$, $W \leq V$ be $T$-invarient subspace. Then,

$$f_{T_W} \text{ divides } f_T.$$

Proof. Let $\beta = \{v_1, \cdots, v_k\}$ be a basis of $W$, and $\bar{\beta} = \{v_1, \cdots, v_k, \cdots, v_n\}$ be a basis of $V$.

Then, $$[T]_{\bar{\beta}} = \left(\begin{array}{c|c} [T_W]_\beta & \cdot \\ \hline O & \cdot \end{array}\right)$$

because for all $1 \leq i \leq k$, $T(v_i) = \sum_{j=1}^{k} a_j v_j$ for some $a_j \in F$.

Now, a characteristic polynomial of $T$:

$$\det([T]_{\bar{\beta}} - tI) = \det([T_W]_\beta - tI) \cdot \det(\cdot - tI)$$

That is, $f_{T_W}$ divides $f_T$.

**Thm.** Let $T: V \to V$ be a linear operator on fin. v.s $V$,

And, $W = \langle \{T^n(v) \mid n = 0, 1, \cdots\} \rangle$ where $v \neq 0$.

Put $k = \dim W$

$$\det \begin{pmatrix} -t & & & & 0 \\ 1 & -t & & & \\ 0 & 1 & \ddots & & \\ \vdots & 0 & \ddots & \ddots & \\ 0 & 0 & \cdots & 1 & -t \end{pmatrix} \quad k$$

$$= (-1)^k t^k - 1$$

1). $\{v, T(v), \cdots, T^{k-1}(v)\}$ is basis of $W$

2). If $a_0 v + a_1 T(v) + \cdots + a_{k-1} T^{k-1}(v) + T^k(v) = 0$,

then $f_{T_W}(t) = (-1)^k (a_0 + a_1 t + \cdots + a_{k-1} t^{k-1} + t^k)$

**Proof.** Let $j \in \mathbb{N}$ be the largest number s.t

$$\beta = \{v, T(v), T^2(v), \cdots, T^{j-1}(v)\}$$

is linearly independent. Put $Z = \langle \beta \rangle$, Then, $\beta$ is basis of $Z$.

Moreover, $\beta \cup \{T(v)\}$ is not linearly independent,

$$T^j(v) = \sum_{i=0}^{j-1} b_i T^i(v). \in Z.$$

Now, for any $w = \sum_{i=0}^{j-1} b_i T^i(v) \in Z$,

$$T(w) = T\left(\sum_{i=0}^{j-1} b_i T^i(v)\right) = \sum_{i=0}^{j-1} b_i T^{i+1}(v) \in Z.$$

Since lemma $W \leq Z$, and $Z \leq W$, because $\beta$ is subset of spanning set of $W$.

To prove 2), if $a_0 v + a_1 T(v) + \cdots + a_{k-1} T^{k-1}(v) + T^k(v) = 0$, then

$$[T_W]_\beta = \begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & \vdots \\ & & \ddots & & \\ 0 & 0 & \cdots & 1 & -a_{k-1} \end{pmatrix}, \quad \text{thus } f_{T_W}(t) = (-1)^k (a_0 + a_1 t + \cdots + a_{k-1} t^{k-1} + t^k).$$

**Lemma.** Let $T: V \to V$ be linear, $W$ be a $T$-cyclic subspace generated by $v \neq 0$.

Then, $W$ is the smallest $T$-invariant subspace containing $v \in V$.

**Proof.** Let $x \in W$ be given, Then, $x$ can be represented by

$$x = \sum_{i=0}^{n} a_i T^i(v)$$

Now, $T(x) = \sum_{i=0}^{n} a_i T^{i+1}(v) \in W$.

And, suppose that $A \leq V$ is $T$-invariant. containing $v \in V$.

Then, $\{v, T(v), T^2(v), \cdots\} \subseteq A$, thus $W \leq A$.

$$\begin{pmatrix} -t & 0 & 0 & \cdots & \vdots & -a_0 \\ 1 & -t & & & \vdots & -a_1 \\ 0 & 1 & \ddots & & \vdots & \vdots \\ \vdots & 0 & \ddots & \ddots & \vdots & \vdots \\ \vdots & \vdots & & \ddots & -t & \vdots \\ 0 & 0 & \cdots & 0 & 1 & -a_{n-1}-t \end{pmatrix} = A_n. \qquad \underbrace{\begin{pmatrix} -t & 0 & & & \\ 1 & -t & & 0 & \\ 0 & 1 & \ddots & & \\ \vdots & 0 & \ddots & \ddots & \\ 0 & 0 & \cdots & 0 \; 1 & -t \end{pmatrix}}_{\det B_n = (-1)^n t^n} = B_n.$$

$$\det A_n = \underline{(-a_{n-1}-t)\cdot \det B_{n-1}}$$
$$+ a_{n-2}\cdot \det B_{n-2}.$$
$$+ a_{n-3}\cdot \det B_{n-3}$$
$$\vdots$$
$$+ a_0 \det B_0$$

[5.4 #3.

$\because \forall v \in E_\lambda, \quad T(v) = \lambda v \in E_\lambda.$

[5.4 #4.

If $T[W] \subseteq W$, then $g(T)[W] \subseteq W$, for any $g(t) \in F[x]$.

proof. Let $g(t) = \sum_{i=0}^{n} a_i t^i$. Then, for any $x \in W$,

$$g(T)(x) = \left(\sum_{i=0}^{n} a_i T^i\right)(x)$$
$$= \sum_{i=0}^{n} a_i T^i(x) \in W, \text{ because } T^i(x) \in W.$$

[5.4 #5.

Let $W_1, W_2 \leq V$ be $T$-invariant.

for any $x \in W_1 \cap W_2$, $T(x) \in W_1$ and $T(x) \in W_2$. $\therefore T(x) \in W_1 \cap W_2$.

[5.4 #13, 14.

Let $T: V \to V$ be a linear operator, $W$ be a $T$-cyclic subspace generated by $v \in V$.

For any $x \in V$,

$x \in W$ iff $\exists g(t) \in F[t]$ s.t $x = g(T)(v)$.

proof. $x \in W \iff x = \sum_{i=0}^{n} a_i T^i(v)$ for some $n \in \mathbb{N}$, $a_i \in F$.

Since for $k = \dim W$, $\{T^0(v), \dots, T^{k-1}(v)\}$ is basis of $W$, clear.

「Sec 5.4 #17

For $A \in M_{n\times n}$, $\dim(\langle\{I, A, A^2, \cdots\}\rangle) \le n$.

proof. Let $f(t) = \det(A - tI)$. Then, $f(A) = 0$.

Since $\deg f(t) = n$, $\{I, A, \cdots, A^n\}$ is linearly dependent,

「Sec 5.4 #18.

Let $A \in M_{n\times n}(F)$ with char. poly.

$$f(t) = (-1)^n t^n + a_{n-1}t^{n-1} + \cdots + a_1 t + a_0.$$

Clearly, $a_0 = \det A$.

「Sec 5.4 #20.

Let $T: V \to V$, and $V = \langle\{T^n(v) \mid n = 0, 1, \cdots\}\rangle$ for some $v \in V$. Then,

A linear map $U: V \to V$ satisfies $T \circ U = U \circ T$

iff

$U = g(T)$ for some $g(t) \in F[t]$

proof. $\Leftarrow$ is trivial,

$\Rightarrow$) Since $U(v) \in V$, $U(v) = g(T)(v)$ for some $g(t) \in F[t]$.

Now, for any $x = \sum a_i T^i(v) \in V$,

$$\begin{aligned} U(x) = U\left(\sum a_i T^i(v)\right) &= \sum a_i T^i(U(v)) \\ &= \sum a_i T^i(g(T)(v)) \\ &= g(T)\left(\sum a_i T^i(v)\right) \\ &= g(T)(x). \quad \blacksquare \end{aligned}$$

#21, 22 is trivial by #20.

Sec 5.4 #23.

$T: V \to V$, $W$: $T$-invariant, $\lambda_1, \cdots, \lambda_k$ distinct eigenvalue. Corresponding $v_i$.

$$v_1 + v_2 + \cdots + v_k \in W \quad \text{iff} \quad v_1, v_2, \cdots, v_k \in W.$$

Proof. For $k=1$, trivial.

For $k \in \mathbb{N}$, suppose it is true.

Now, for $v_1, \cdots, v_{k+1}$, suppose:

$$v_1 + v_2 + \cdots + v_{k+1} \in W.$$

Then, $\sum_{i=1}^{k+1} T(v_i) \in W$, and

$$= \sum_{i=1}^{k+1} \lambda_i v_i.$$

If $\lambda_{k+1} \neq 0$, then $\lambda_{k+1}(\sum^{k+1} v_i) \in W$, this implies

$$\sum_{i=1}^{k+1} \lambda_i v_i - \lambda_{k+1}\left(\sum_{i=1}^{k+1} v_i\right)$$

$$= \sum_{i=1}^{k+1} (\lambda_i - \lambda_{k+1}) v_i$$

$$= \sum_{i=1}^{k} (\lambda_i - \lambda_{k+1}) v_i \in W \Rightarrow v_1, \cdots, v_k \in W,$$

Thus $v_{k+1} \in W$.

If $\lambda_{k+1} = 0$, then clearly $v_1, \cdots, v_k \in W$, obtain same conclusion.

Sec 5.4 #24 ?

Let $T: V \to V$ be a diagonalizable. Then, for any $T$-invariant $W$,

$$T_W \text{ is diagonalizable}$$

Proof. Since $T$ is diagonalizable, $V = \bigoplus_{i=1}^{n} E_{\lambda_i}$, where $\lambda_i$ is distinct eigenvalue of $T$.

Now, for any $x \in W$, $x = \sum_{i=1}^{n} a_i v_i$ as $\begin{matrix} a_i \in F, \\ v_i \in E_{\lambda_i} \end{matrix}$, and unique representation.

Since above Lemma, all $v_i \in W$. That is,

$$W = \bigoplus_{i=1}^{n} (E_{\lambda_i} \cap W)$$

Sec 5.4 #25.

Let $T, U: V \to V$ be diagonalizable.

If $T \circ U = U \circ T$, then $T, U$ simultaneously diagonalizable.

Proof. Let $\lambda \in F$ be a non-zero eigenvalue of $T$. For $E_\lambda = \{v \in V \mid T(v) = \lambda v\}$,

Let $x \in E_\lambda$. Then, $U(x) = U(\lambda^{-1} T(x)) = \lambda^{-1}(T \circ U)(x)$

$\iff \lambda U(x) = T(U(x))$. That is, $U(x) \in E_\lambda$.

If $\lambda = 0$, then $E_\lambda = \ker T$.

Let $x \in \ker T$. Then, $T(U(x)) = U(T(x)) = 0$. $\therefore U(x) \in \ker T$.

step 1

In sum, $E_\lambda$ is $U$-invariant.

Now, $U_{E_\lambda}$ is diagonalizable, put $\beta_\lambda$ is basis of $E_\lambda$ s.t. $[U_{E_\lambda}]_{\beta_\lambda}$ is diagonal

$\beta = \bigcup_\lambda \beta_\lambda$ is basis of $V$ consisted by eigenvectors of $T$.

and $[U]_\beta$ is diagonal.

$T$ diagonalizable

$$\Rightarrow \underset{\text{span}\uparrow}{V} = \underset{\uparrow}{E_{\lambda_1}} \oplus \cdots \oplus \underset{\uparrow}{E_{\lambda_k}}$$

$$\beta = \beta_1 \cup \cdots \cup \beta_k.$$

Sec 5.4, #26.

Let $T: F^n \to F^n$ be a linear, having $n$ numbers distince eigenvalues.

Then, $F^n$ is $T$-cyclic space.

proof. Let $\lambda_1, \cdots, \lambda_n$ be eigenvectors of $T$.

pick $v_i$ be eigenvector corresponding to $\lambda_i$.

Let $v = \sum_{i=1}^{n} v_i$, $W = \langle \{v, T(v), \cdots\} \rangle$.

Then, $W$ must contains of $v_i$'s, thus $W = \oplus E_{\lambda_i} = V$.

$\ulcorner$ $T: V \to V$, $W$: $T$-invariant

$\overline{T}: V/W \to V/W : v+W \mapsto T(v)+W$.

$\eta: V \to V/W ; v \mapsto v+W$.

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "\eta"'] & V \arrow[d, "\eta"] \\ V/W \arrow[r, "\overline{T}"] & V/W. \end{tikzcd}$$

$\ulcorner$ Thm. $f_T(t) = f_{\overline{T}}(t) \cdot f_{T_W}(t)$.

Proof. Let $\gamma = \{v_1, \cdots, v_k\}$ be a basis of $W$,

$\beta = \{v_1, \cdots, v_k, \cdots, v_n\}$ be a basis of $V$.

Then, $\alpha = \{v_{k+1}+W, \cdots, v_n+W\}$ is basis of $V/W$.

Now,

$$[T]_\beta = \left(\begin{array}{c|c} [T]_\gamma & * \\ \hline O & [\overline{T}]_\alpha \end{array}\right).$$

「Sec 5.4 #32.

Let $T: V \to V$ be linear operator, $W_1, \cdots, W_k$ be $T$-invariant subspace.
Then, $W_1 + \cdots + W_k$ is $T$-invariant subspace.

Proof. Let $x_1 + \cdots + x_k \in \sum_{i=1}^{k} W_i$ be given. Then,

$$T(x_1 + \cdots + x_k) = \sum_{i=1}^{k} T(x_i) \in \sum_{i=1}^{k} W_i, \text{ since each } T(x_i) \in W_i.$$

「Sec 5.4 #33.

Let $T: V \to V$ be a linear, $V$ is finite-dimensional. $V = W_1 \oplus W_2$ for some $T$-invariantes $W_1, W_2$.
Let $\beta_1, \beta_2$ be basis of $W_1, W_2$, respectively. Put $\beta = \beta_1 \cup \beta_2$
Then,

$$[T]_\beta = [T_{W_1}]_{\beta_1} \oplus [T_{W_2}]_{\beta_2}.$$

Proof. Since $V$ is direct sum of $W_1$ and $W_2$, $\beta_1 \cap \beta_2 = \emptyset$.
Denote $\beta_1 = \{v_1, \cdots, v_n\}$, $\beta_2 = \{w_1, \cdots, w_m\}$.

$$T(v_i) = T_{W_1}(v_i) \in W_1, \quad T(w_i) = T_{W_2}(w_i) \in W_2.$$

Sec 5.4 #35.

Let $T: V \to V$ be a linear operator on finite dimensional $V$.

$T$ is diagonalizable iff $V = \bigoplus_{i=1}^{n} W_i$ where $\dim W_i = 1$, $W_i$ is $T$-invariant.

Proof. Suppose that $T$ is diagonalizable. Then, There exists a basis $\beta$ s.t. $[T]_\beta$ is diagonal. Then, for each $v_i \in \beta$, $\langle v_i \rangle$ is $T$-invariant, since

$$T(av_i) = a\lambda_i v_i \in \langle v_i \rangle, \text{ for any } a \in F.$$

Now, $V = \bigoplus \langle v_i \rangle$.

$$A = \begin{pmatrix} 1 & 2 & \cdots & n \\ n+1 & n+2 & & 2n \\ 2n+1 & \vdots & \ddots & \vdots \\ \vdots & & & \\ n^2-n+1 & n^2-n+2 & & n^2 \end{pmatrix}$$

Since $A^{[2]} - A^{[1]} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = 1 \cdot_{,}$

$kB + A^{[1]} = A^{[k+1]}$. And, $\{A^{[1]}, A^{[2]}\}$ linearly independent,

thus $\operatorname{rank} A = 2$.

And, put $V_1 = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$, $V_2 = \begin{pmatrix} 1 \\ 2 \\ \vdots \\ n \end{pmatrix}$. Then, for any $a_1, a_2 \in F$,

$$A(a_1v_1 + a_2v_2)$$

$$= a_1 \begin{pmatrix} \sum_{k=1}^{n} k \\ n^2 + \sum_{k=1}^{n} k \\ 2n^2 + \sum_{k=1}^{n} k \\ \vdots \\ n^3 + \sum_{k=1}^{n} k \end{pmatrix} + a_2 \begin{pmatrix} \sum_{k=1}^{n} k^2 \\ n \cdot \left(\sum_{k=1}^{n} k\right) + \sum_{k=1}^{n} k^2 \\ 2n \cdot \left(\sum_{k=1}^{n} k\right) + \sum_{k=1}^{n} k^2 \\ \vdots \\ n^2 \cdot \left(\sum_{k=1}^{n} k\right) + \sum_{k=1}^{n} k^2 \end{pmatrix} \quad a_1 \begin{pmatrix} \frac{1}{2}n(n+1) \\ n^2 + \frac{1}{2}n(n+1) \\ 2n^2 + \frac{1}{2}n(n+1) \\ \vdots \\ n^3 + \frac{1}{2}n(n+1) \end{pmatrix} + a_2 \Bigg($$

$A[\langle v_1, v_2 \rangle]$ has rank 2.

$Av_1 = n^2 v_2 + \sum_{k=1}^{n} k \cdot v_1$

$Av_2 = \left(\sum_{k=1}^{n} k^2\right) v_1 +$

$$A = \begin{pmatrix} 0 \cdot n \cdot v_1^t v_2^t \\ 1 \cdot n \cdot v_1^t + v_2^t \\ 2 \cdot n \cdot v_1^t + v_2^t \\ \vdots \\ (n-1) \cdot n \cdot v_1^t + v_2^t \end{pmatrix}$$